Thesis Figure Evaluation — umc_thesis

Generated: 27 May 2026 | 82 figures in `outputs/figures/` + 16 in `outputs/old_figures/`

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1. Full Inventory

Below is every figure, grouped by type. ■ = keep for thesis | ■ = needs work before keeping | ■ = archive/delete.

1a. EDA / Data Description (3 figures)

File	Content	Status
`01_component_distribution.png`	Stone count per primary class, stacked by purity	
`02_threshold_sweep.png`	Pure/mixed balance across purity thresholds	
`03_total_presence.png`	Component occurrence across all positions	

1b. Architecture Diagrams (2 figures)

File	Content	Status
`arch_model_a.png`	Model A – binary classifier (ResNet18 + head)	
`arch_model_b.png`	Model B – parallel vs simple composition heads	

1c. ROC Curves (20 figures)

These come in two series: **cross-validation** (no suffix) and **test set** (`_test`).

Files (CV / Test)	Model	Thesis relevance
`roc_comparison.png` / `_test`	t100 vs t90 vs random baseline	**Key figure**
`roc_curve_resnet18.png` / `_test`	ResNet18 binary	
`roc_curve_efficientnet.png` / `_test`	EfficientNet binary	(keep if ablation)
`roc_curve_random.png` / `_test`	Random baseline	(already in comparison)
`roc_curve_t90.png` / `_test`	t90 threshold	(already in comparison)
`roc_curve_t100.png` / `_test`	t100 threshold	(already in comparison)
`roc_curve_random_resnet18_reg.png` / `_test`	Random baseline on composition	(intermediate)
`roc_curve_t90_resnet18_reg.png` / `_test`	t90 on composition	(intermediate)
`roc_curve_t100_resnet18_reg.png` / `_test`	t100 on composition	(intermediate)

Files (CV / Test)	Model	Thesis relevance
`roc_curve_wd001_lrlow.png` / `_test`	Hyperparam experiment	(intermediate)

1d. Confusion Matrices (26 figures)

Two families: **binary 2×2** (Model A) and **multiclass 5×5** (Model C). Plus composition-level "primary confusion" figures.

Files (CV / Test)	Model	Thesis relevance
`confusion_matrix_resnet18.png` / `_test`	ResNet18 binary	
`confusion_matrix_efficientnet.png` / `_test`	EfficientNet binary	(if ablation shown)
`confusion_matrix_multiclass_resnet18_reg.png` / `_test`	Multiclass Model C	**Key figure**
`parallel_primary_confusion.png` / `_test`	Parallel composition, dominant class	
`simple_primary_confusion.png` / `_test`	Simple composition, dominant class	
`simple_resnet18_reg_primary_confusion.png` / `_test`	Simple+ResNet18 composition	
`confusion_matrix_t90.png` / `_test`	t90 threshold	(intermediate)
`confusion_matrix_t100.png` / `_test`	t100 threshold	(intermediate)
`confusion_matrix_random.png` / `_test`	Random baseline	(intermediate)
`confusion_matrix_multiclass.png` / `_test`	Multiclass vanilla	(intermediate)
`confusion_matrix_multiclass_random.png` / `_test`	Multiclass random	(intermediate)
`confusion_matrix_*.png` / `_test` for `wd001_lrlow`, `random_resnet18_reg`, etc.	Hyperparam experiments	

1e. Training Curves (12 figures)

File	Model	Thesis relevance
`training_curves_resnet18.png`	Model A binary (ResNet18)	(representative)
`simple_resnet18_reg_training_curves.png`	Model B composition	
`training_curves_efficientnet.png`	EfficientNet	(if ablation)
`training_curves_multiclass_resnet18_reg.png`	Model C multiclass	
`training_curves_t90.png`	t90 experiment	
`training_curves_t100.png`	t100 experiment	

File	Model	Thesis relevance
`training_curves_random.png`	Random baseline	
`training_curves_random_resnet18_reg.png`	Random composition	
`training_curves_multiclass.png`	Multiclass vanilla	
`training_curves_multiclass_random.png`	Multiclass random	
`training_curves_t90_resnet18_reg.png`	t90 composition	
`training_curves_wd001_lrlow.png`	Hyperparam experiment	

1f. Composition Regression Figures (14 figures)

Files (CV / Test)	Content	Thesis relevance
`parallel_composition_scatter.png` / `_test`	Parallel model: predicted vs true per component	
`simple_composition_scatter.png` / `_test`	Simple model: predicted vs true per component	
`simple_resnet18_reg_composition_scatter.png` / `_test`	Simple+ResNet18	
`parallel_mae_bars.png` / `_test`	Per-class MAE bars, parallel model	
`simple_mae_bars.png` / `_test`	Per-class MAE bars, simple model	
`simple_resnet18_reg_mae_bars.png`	**MISSING** — exists for scatter but not MAE bars	regenerate if needed

1g. Explainability (9 figures)

File	Content	Thesis relevance
`gradcam/summary_grid.png`	4 outcomes × 3 stone examples with CAM overlay	**Key figure**
`gradcam/true_positives.png`	TP examples	(appendix)
`gradcam/true_negatives.png`	TN examples	(appendix)
`gradcam/false_positives.png`	FP examples	(appendix)
`gradcam/false_negatives.png`	FN examples	(appendix)
`embeddings/tsne_binary_resnet18.png`	t-SNE, binary features, ResNet18	
`embeddings/tsne_binary_efficientnet.png`	t-SNE, binary features, EfficientNet	
`embeddings/tsne_multiclass_resnet18.png`	t-SNE, multiclass features	
`embeddings/tsne_multiclass_efficientnet.png`	t-SNE, multiclass features, EfficientNet	

1h. old_figures/ (16 figures)

All figures here are superseded by figures in `figures/`. **Archive or delete the entire folder.**

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2. Recommended Thesis Figure Set (~25 figures)

This is a lean, complete set covering every thesis section. Rename them to a clean, sequential scheme for the final document.

Thesis section	Figure file(s) to use
Methods — Data	`01_component_distribution.png`, `02_threshold_sweep.png`
Methods — Architecture	`arch_model_a.png`, `arch_model_b.png` (after fix)
Results — Model A binary	`confusion_matrix_test_resnet18.png`, `roc_comparison_test.png`, `training_curves_resnet18.png`
Results — Model B composition	`simple_composition_scatter_test.png`, `parallel_composition_scatter_test.png`, `simple_mae_bars_test.png`, `parallel_mae_bars_test.png`, `simple_primary_confusion_test.png`, `parallel_primary_confusion_test.png`, `simple_resnet18_reg_training_curves.png`
Results — Model C multiclass	`confusion_matrix_test_multiclass_resnet18_reg.png`, `training_curves_multiclass_resnet18_reg.png`
Analysis — Explainability	`gradcam/summary_grid.png`, `embeddings/tsne_binary_resnet18.png`, `embeddings/tsne_multiclass_resnet18.png`
Appendix — GradCAM detail	`gradcam/true_positives.png`, `gradcam/false_positives.png`, `gradcam/false_negatives.png`
Appendix — EDA detail	`03_total_presence.png`

Total: ~23 figures. The remaining ~59 are intermediate experiments — move them to a `figures/intermediate/` subfolder, don't delete (you'll need them if reviewers ask questions).

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3. Consistency Issues (Fix Before Submission)

3.1 Critical — Architecture diagram B is unreadable

`arch_model_b.png` shows a side-by-side comparison of Parallel vs Simple heads, but the text is too small to read when printed at A4. Two options:

- **Option A (preferred):** Split into two separate figures — `arch_model_b_parallel.png` and `arch_model_b_simple.png` — and use the same style/size as `arch_model_a.png`.
- **Option B:** Double the figure width (≥1600 px) and increase all font sizes to match Model A.

3.2 Critical — Confusion matrices show raw counts only, no percentages

This is a standard issue that examiners notice. Raw counts are hard to compare when row totals differ (e.g., CHP has many more stones than MAP). Fix: add percentage annotations inside each cell, e.g. `19\n(83%)`. The `text_kw` argument in `sklearn.metrics.ConfusionMatrixDisplay` or a custom `ax.text()` call handles this.

For binary matrices: also add per-class recall (sensitivity/specificity) as annotation on the right side.

3.3 Important — AUC ± 0.000 in ROC curve legends is misleading

In `roc_comparison.png` and `roc_comparison_test.png`, all models show `AUC = X.XXX ± 0.000`. This happens when the ROC curve is computed on the full aggregated test fold rather than per-fold. Either:

- Compute AUC per fold and report mean ± std (the biologically meaningful number), or
- Remove the `± 0.000` and just show `AUC = X.XXX`.

As-is, a reader will think you computed fold-level AUCs and got zero variance, which raises suspicion.

3.4 Important — No unified color palette

Each figure type uses a different color system:

Figure type	Colors currently used
EDA bar charts	Green + salmon
Binary confusion matrix	Blue sequential colormap
Training curves	Matplotlib default (blue/red)
MAE bars	Steel blue
Composition scatter	Default blue
t-SNE	Blue/red (pure/mixed)
GradCAM	Viridis/hot + grayscale

Suggested fix: Define a small palette at the top of your visualization scripts and import it everywhere:

```
COLORS = {
"pure": "#2166AC", # dark blue
"mixed": "#D6604D", # muted red
"train": "#4393C3", # medium blue
"val": "#D6604D", # muted red
"accent": "#F4A582", # light orange
"neutral": "#92C5DE", # light blue
}
```

This doesn't require redesigning every figure — just updating the 3–4 colors that recur most (pure/mixed, train/val).

3.5 Important — Title format is slightly inconsistent

All titles use em-dashes (—), which is good. But the structure varies:

- `"Stone count per primary component (threshold=100%)"` — descriptive inline
- `"Confusion matrix (aggregated across folds) — resnet18"` — descriptor + model appended
- `"Confusion matrix — primary component (cross-validation) — resnet18_reg"` — different order
- `"t-SNE — backbone features colored by pure/mixed (fold 0) — resnet18"` — very long

Suggested standard:

```
[Figure type] — [What it shows] — [Model name], [Split]
```

Examples:

```
Confusion matrix — primary component — ResNet18, test set
ROC curve comparison — 100% / 90% purity thresholds — test set
Training curves — binary classifier — ResNet18
t-SNE — backbone feature space — ResNet18 (binary)
```

Apply this consistently and your figures will look like they belong in the same document.

3.6 Minor — Training curves show only 1 fold

`training_curves_resnet18.png` shows fold 0 only. For a 5-fold setup, either:

- Plot all 5 folds in light grey + mean in bold (standard), or
- Plot fold 0 explicitly and add a note: *"Representative fold shown; all folds shown in Appendix"*.

The composition training curve (`simple_resnet18_reg_training_curves.png`) is actually very well done — the two-panel layout showing both loss and MAE/accuracy simultaneously is the best training curve in the set. Use this as the template for the others.

3.7 Minor — GradCAM summary grid is too small

`gradcam/summary_grid.png` is the most important explainability figure, but the stones and annotations are barely readable. Increase DPI (≥200) or figure size (at least 14×10 inches) when regenerating. The individual outcome files (FP, FN, etc.) are better-sized.

3.8 Minor — Composition scatter: inconsistent y-axis ranges

In `simple_composition_scatter.png`, each subplot has a different y-axis range (some go to 0.7, others to 1.0). This makes visual comparison harder. Fix: standardize all axes to `[0, 1]` for both x and y. Also consider adding a diagonal identity line more prominently (it's already there but thin).

3.9 Minor — t-SNE is from fold 0 only, not noted clearly

The title says `(fold 0)` which is good, but without context a reader might wonder why not all folds. Add a short caption note or regenerate on all training data.

3.10 Minor — `simple_resnet18_reg_mae_bars.png` is missing

The `simple_resnet18_reg` model has composition scatter and primary confusion figures, but no MAE bars. Either add this figure or drop the model from comparison (for consistency, every model you compare should have the same figure set).

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4. Per-Figure-Type Quality Summary

EDA figures (01–03)

Strong. Clean, well-annotated, counts visible on bars. One issue: `01_component_distribution.png` uses a **green/salmon** color scheme for Pure/Near-pure/Mixed, while nothing else in the project uses those colors. Since this is an EDA figure (not a model result), it's acceptable to keep the dedicated palette — but make sure the same colors appear in `02_threshold_sweep.png` too (currently `02` uses green/salmon for bar chart but plain blue/red for line chart — unify these).

Architecture diagrams

Model A: excellent. Clear layout, readable text, good color coding by layer type.

Model B: needs work (see §3.1). The concept is good (side-by-side comparison is useful), but the execution is too compressed. The hyperparameter box at the bottom is a nice touch — keep it.

Confusion matrices

Binary (2×2): Clean but raw counts only. Normalization fix is essential (§3.2). The "pure"/"mixed" ordering (pure first) is good and should be kept consistent.

Multiclass (5×5): Good layout. The class ordering (CaOx, CHP, UA, MAP, CYS) is logical. Consider reordering by decreasing frequency if not already done. The colormap (Blues sequential) works well for this.

Composition-level "primary confusion": These are 5×5 matrices that use a different title format. Bring title format in line with the standard (§3.5).

ROC curves

Individual model curves: Serviceable, but stepped ROC curves look rough on small datasets. No smoothing is correct statistically, but be aware this looks less polished than high-N papers. Include a note in the caption.

Comparison figure (`roc_comparison_test`): This is your most important performance figure. Fix the `± 0.000` issue (§3.3). Consider adding shaded bands for fold-level variation. The three-color comparison (red/blue/green) is legible.

Training curves

The binary model curve is concerning scientifically: validation loss barely moves (stays ~0.69–0.70 throughout), suggesting the model is not learning much from the data. This is worth discussing explicitly in the thesis — don't hide it. The plot is technically correct, just shows a hard problem.

The composition curve (`simple_resnet18_reg`) is the best in the set: dual panel, shows the MAE trajectory alongside loss, good epoch count. Use this as the template.

Composition scatter / MAE bars

Scatter: Clean but the per-panel axes vary (fix: §3.8). No regression line, which is fine.

MAE bars: Excellent. Error bars present (±1 SD), overall reference line shown, clean labels. One of the best-quality figures in the set. Just align axis label ("MAE") style with other figures.

GradCAM / Explainability

Summary grid: The concept is excellent — 4 outcomes × 3 photos in one figure tells a compelling story. But needs to be larger (§3.7). The dark background works well for CAM overlays.

Individual FP/FN grids: Good for appendix. The annotation format (stone ID + prediction confidence) is informative.

t-SNE: Clean, uncluttered. The overlap between Pure/Mixed in feature space is visible and matches the model performance story — this is scientifically honest and good to show.

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5. Action Checklist (Priority Order)

- **Immediate:** Archive `outputs/old_figures/` — do not include in thesis
- **Immediate:** Fix `arch_model_b.png` — text unreadable at print size
- **Immediate:** Add percentage annotations to all confusion matrices
- **High:** Fix `AUC ± 0.000` in ROC legend — compute per-fold AUC std

- **High:** Create `outputs/figures/thesis_final/` folder with the ~23 recommended figures
- **High:** Rename the ~23 final figures to sequential scheme (e.g. `fig01_...`, `fig02_...`)
- **Medium:** Standardize titles to consistent format (§3.5)
- **Medium:** Unify color palette for pure/mixed and train/val across scripts
- **Medium:** Increase GradCAM summary grid DPI/size
- **Medium:** Fix composition scatter y-axis to [0, 1] for all panels
- **Low:** Add multi-fold display to binary training curves
- **Low:** Add `simple_resnet18_reg_mae_bars.png` or drop model from comparison

End of evaluation. Figure counts: 82 active + 16 archived = 98 total. Recommended thesis set: ~23.